We've so far assumed that space is shaped like a video game screen, but the story has many of the same features for the other possibilities. For example, if the data ultimately show that the shape of space is spherical, then, as we go ever farther back in time, the size of the sphere gets ever smaller, the universe gets ever hotter and denser, and at time zero we encounter some kind of big bang beginning. Drawing an illustration analogous to Figure 8.10 is challenging since spheres don't neatly stack one next to the other (you can, for example, imagine a "spherical loaf" with each slice being a sphere that surrounds the previous), but aside from the graphic complications, the physics is largely the same.

The cases of infinite flat space and of infinite saddle-shaped space also share many features with the two shapes already discussed, but they do differ in one essential way. Take a look at Figure 8.11, in which the slices represent flat space that goes on forever (of which we can show only a portion, of course). As you look at ever earlier times, space shrinks; galaxies get closer and closer together the farther back you look in Figure 8.11b. However, the overall size of space stays the same. Why? Well, infinity is a funny thing. If space is infinite and you shrink all distances by a factor of two, the size of space becomes half of infinity, and that is still infinite. So although everything gets closer together and the densities get ever higher as you head further back in time, the overall size of the universe stays infinite; things get dense everywhere on an infinite spatial expanse. This yields a rather different image of the big bang.

Normally, we imagine the universe began as a dot, roughly as in Figure 8.10, in which there is no exterior space or time. Then, from some kind of eruption, space and time unfurled from their compressed form and the expanding universe took flight. But if the universe is spatially infinite, *there was already an infinite spatial expanse at the moment of the big bang.* At this initial moment, the energy density soared and an incomparably large temperature was reached, but these extreme conditions existed everywhere, not just at one single point. In this setting, the big bang did not take place at one point; instead, the big bang eruption took place *everywhere* on the infinite expanse. Comparing this to the conventional single-dot beginning, it is as though there were many big bangs, one at each point on the infinite spatial expanse. After the bang, space swelled, but its overall size didn't increase since something